

Advika Nagool

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Data Science student exploring Machine Learning, AI, and Quantum Computing, with experience building AI systems, applied research, and data-driven pipelines. Research accepted at IICCCS and presented at Comsys 2026. National gold medallist in archery, bringing precision, discipline, and structured problem-solving to technology.

EDUCATION

Degree	Institute / Board	CGPA / %	Year
MBA + BTech (Data Science)	Mukesh Patel School of Technology Management & Engineering, Mumbai	9.4 / 10 CGPA	2024– Present
Class XII	Pace Junior Science College, Mumbai	88.50%	2022–2024
Class X	I.E.S Orion, Mumbai	95.40%	2010–2022

TECHNICAL SKILLS

Languages: Python, C++, C-family (Java, Kotlin), R, JavaScript, HTML, CSS
Scientific Computing & Machine Learning: NumPy, Pandas, SciPy, Scikit-learn, Matplotlib, PyTorch, TensorFlow
Data & Analysis: MySQL, Excel, Power BI, Tableau
Backend & AI APIs: FastAPI, Django, LangChain, LangGraph, Firebase, Claude API, Groq API
Cloud, DevOps & Tools: Docker, AWS, Git/GitHub, VS Code, Android Studio, Jupyter

WORK EXPERIENCE

Marketing Makhni | Research & Development Intern — **Mar 2026 – Aug 2026**

- Conducted competitive analysis for 10+ companies, gathering and structuring market, competitor, and pricing data through web research, scraping, and client-side data acquisition to support business and strategic decision-making.
- Led client-side research projects from requirement gathering to final delivery, coordinating with stakeholders, managing data acquisition workflows, and ensuring timely delivery of analysis-ready datasets and project outputs.

Adihuman AI | AI Research Intern — **Dec 2025 – Mar 2026**

- Developed a structured evaluation rubric for GAIA Level 3 tasks, covering multiple failure modes and enabling consistent assessment of complex AI agent outputs through clearly defined evaluation criteria.
- Evaluated multi-step LangGraph conversational workflows across 50+ edge cases, improving consistency in tone, voice, and emotional register through prompt frameworks with explicit constraints.

PROJECTS

Multi-Agent Market Analysis System — Evidence-Gated Brand Recommendations — **2026**

- Built a four-agent Python system (Research → Analysis → Investigation → Control) that answers brand-attention questions from live APIs (Wikipedia, Media Cloud, YouTube, Wayback), detecting change points with Welch/Poisson tests under FDR correction and clustering them into cross-source events.
- Added a deterministic citation gate that rejects any LLM claim without a matching retrieved record, so the system returns "insufficient evidence" instead of inventing an explanation; verified with 600+ tests.

Stoch Claim — Probabilistic Sequence Modelling & Decision Pipeline — **2026**

- Built a decision pipeline that routes a claim to the right action in under 7 seconds, using Hidden Markov Models for latent-state sequence estimation and Markov Decision Processes for value-iteration routing over 1,000 records.
- Combined stochastic modelling (HMM, MDP, Gambler's Ruin) with LLM-generated explanations (Llama-3.3-70B via Groq) for cost-optimal routing across five action classes with full audit trails.

Braille Bridge — Noise-Robust Tactile Signal Processing & Recognition — **2025**

- Built a word-level tactile Braille recognition pipeline, deployed via Gradio and React, that reads raw sensor spike trains from the iCub taxel-array dataset and predicts character sequences without frame-level alignment.
- Cleaned raw sensor input with a 1D convolutional denoising autoencoder ahead of a CNN-LSTM with CTC decoding, and isolated the denoising stage's contribution through ablation studies.

RESEARCH & PUBLICATIONS

Multimodal DistilBERT–Acoustic Fusion for Emotional State Classification in Naturalistic High-Stress Speech: A Case Study on Formula 1 Team Radio Communication — **2026**

- **Accepted at IICCCS.** Built a multimodal DistilBERT–acoustic fusion model with focal loss and EDA that classifies Calm, Frustrated and High Stress states in Formula 1 team radio speech, improving accuracy from 56% to 68% and macro-F1 from 0.467 to 0.590 over a frozen-backbone baseline.

A Hybrid LLM Framework for Emission-Constrained Supply Chains with Renewable-Aware RL — **2026**

- **Presented at Comsys 2026.** Co-authored a hybrid framework that cut lifecycle CO_2 by 32.9% vs. a cost-minimising plan (+11% cost), combining LSTM/GRU forecasting, NSGA-II optimisation and PPO control, with an LLM layer explaining each stage's results to decision-makers.

VOLUNTEERING

Bhavyata Foundation | Volunteer — **Summer 2025**

- Developed educational materials and bilingual learning resources for early childhood programs, while supporting environmental awareness initiatives through event design and documentation.

CERTIFICATIONS, ATHLETICS & LEADERSHIP

- **Certifications:** Supervised Machine Learning (DeepLearning.AI) · Qiskit Global Summer School 2026 (Quantum Excellence) · Polarizer (2025) :2nd Place Paper Presentation · Techskript (2025): 2nd Place Paper Presentation · AWS Cloud Technical Essentials (AWS)
- **Athletics:** National Gold Medallist — Archery · National Level — Rifle Shooting · School Sports Captain (2020–21)
- **Leadership:** IET, MPSTME — Core Member (2025–2026) · Findrome, MPSTME — Executive (2024–25)